Quiz 1: 25%. Quiz 2: 25%. lnt@ee.iitm.ac.in

endsem: 40%. Tutorial: 10%. A.V. Oppenheim

A signal is a function $f(x)$ $\quad f: R \to C$

- x is independent variable → time, frequency, position
- $f(x)$ is the dependent value → Amplitude, phase
- can visualize signal through a plot

Systems: A box that possesses an input signal (→ voltage signal) and produces an output signal. → signal without DC component

S: f : f (function input, function output)

Both signals and systems are well defined

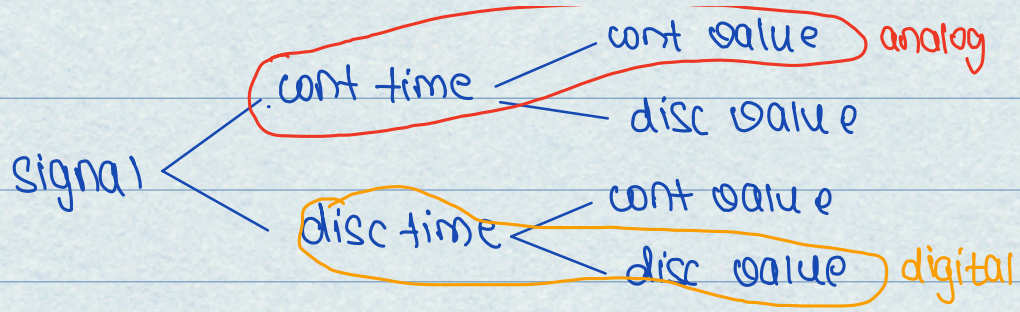
Signal Analysis:

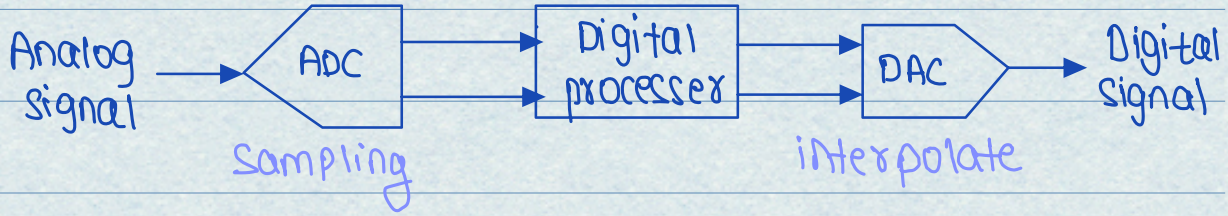
Analytical representation:

$x(t) = \cos t \;,\; t+3$

a function $x(t)$	basically a sequence $x[n]$
Continuous-time signal	Discrete time signal
values of independent variable (t) is from a continuum	independent variables takes only discrete value (dependent can be from continuum)
audio signals, voltage/ current signals	a sampled signal, everyday's temperature

Analog	Digital
continuous value continuous time	discrete value discrete time
amplitude can take values from continuum	discrete signal is further discretized (quantized) in amplitude
can't be stored efficiently, but generated and processed by analog circuits	can be stored and processes by a computer
real life signals (wired, wireless media)	processed in computer / micro-controller





Examples of continuous-time signal:

- $x(t) = A\sin(\omega t + \phi)$ (phase)
  - ω: Angular frequency

  linear frequency $= \omega/2\pi = f$

  period $= \frac{2\pi}{\omega} = T$

- Step and ramp

  $u(t)$: unit step function $= \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$

  const value from time of observation

  (voltage across battery)

  $r(t)$: $\begin{cases} t & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$

  $r(t) = \int_0^t u(\tau)\,d\tau$ (current in inductor)

- $rect(t) = \begin{cases} 1 & \text{if } -1/2 \le t \le 1/2 \\ 0 & \text{otherwise} \end{cases}$ $\quad rect(t) = u(t+1/2) - u(t-1/2)$

Obtained by combination of step signals (voltage signals from USB port)

- $tri(t) = \begin{cases} 1-|t| & \text{if } -1 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$

Obtained by integrating some combination of rectangular pulses

Examples of discrete signals:

- $x[n] = A\sin(\omega n + \phi) = A\sin(2\pi f n + \phi)$ (phase: ϕ; ω: angular frequency)

linear frequency $= \omega/2\pi = f$

period → defined only if f is a rational number

$f = k/m$ when $n = m \Rightarrow 2\pi \frac{k}{m} \times m = 2\pi k$

- Step and ramp

$u[n]$: unit step function $= \begin{cases} 1 & \text{if } n \ge 0 \\ 0 & \text{otherwise} \end{cases}$

const value from time of observation

$$r[n] = \begin{cases} n & \text{if } n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$y[n] = \sum_{k=1}^{n} u[k] \qquad u[n] = r[n+1] - r[n]$$

- $\text{rect}[n] = \begin{cases} 1 & \text{if } -N/2 \leq n \leq N/2 \quad (N \to \text{even}) \\ 0 & \text{otherwise} \end{cases}$

Obtained by combination of step signals

non-zero duration of N samples

- $\text{tri}[n] = \begin{cases} N - |n| & \text{if } -1 \leq n \leq 1 \\ 0 & \text{otherwise} \end{cases}$

Obtained by summing some combination of rectangular pulses

Time Shifting and scaling: forward reverse

- delay / advance $x(t-T)$ / $x(t+T)$

(TV transmission delays, Audio timing adjustments)

- $x(at)$ → speeding-up / slowing down

  $|a|<1$ → expand (slower)

  $|a|>1$ → compress (faster)

  Time reversal: $a<0$

  anchored at $t=0$

- $x(at+b) = y(t)$

  $x(t+b) = x_1(t)$ ✓ $\quad x_1(at) = y(t)$

  $x(at) = x_1(t)$ ✗ $\quad x_1(t+b) = y(t) = x_1\left(t+\frac{b}{a}\right)$ ✓

- $y = x(f_2(f_1(t)))$ → output given, to find input

  ⟹ reverse operation: $x(f_2^{-1}(f_1^{-1}(f_1(f_2(t)))))$

Addition and subtraction of signals:

- Linear combination: $z(t) = Ax(t) + By(t)$
- Signum $(t) = 2u(t) - 1 = u(t) - u(-t)$
- rect $(t) = u(t+1/2) - u(t-1/2)$

- Signals with magnitude and phase:
- Impedance of a component (Z) = resistance + reactance

$V = iZ$

- resistance: opposes flow of current along voltage (in phase)
- reactance: opposes rate of change of current / voltage (out of phase)
- Capacitance: $i = C\frac{dV}{dt}$ opposes change of voltage (voltage lags)
- Inductance: $V = L\frac{di}{dt}$ opposes change of current (current lags)

Even / odd Signal:

- even: $x(t) = x(-t)$ symmetric about $t=0$
- odd: $x(t) = -x(-t)$ symmetric about origin

- $x(t) = \underbrace{\frac{1}{2}[x(t)+x(-t)]}_{\text{even}} + \underbrace{\frac{1}{2}[x(t)-x(-t)]}_{\text{odd}}$

orthogonal to one another

- $\int_{-\infty}^{\infty} x^*(t)\, y(t) = \int_{-\infty}^{\infty} x(t)\, y^*(t) = 0$: orthogonal signals

- **Orthogonal signals** → equivalent to dot product for continuous time signals
  - → for real valued signals, $\int_{-\infty}^{\infty} x(t)y(t)\,dt = 0$
  - $\cos(\omega t)$, $\cos(\theta t)$ → orthogonal if $\omega \neq \theta$
  - $\sin(\omega t)$, $\cos(\theta t)$ → orthogonal
  - $e^{j\omega t}$, $e^{-j\omega t}$ → not orthogonal

Conjugate symmetry:

- $x(t) = x^*(-t)$ → conjugate symmetric
- $x(t) = -x^*(-t)$ → conjugate skew symmetric
- $x(t) = \frac{1}{2}[\underbrace{x(t) + x^*(-t)}_{CS}] + \frac{1}{2}[\underbrace{x(t) - x^*(-t)}_{CSS}]$

- Conjugate symmetric / skew symmetric doesn't affect area (and so, orthogonality)

- for real signals, CS = even and CSS = odd
- conjugate symmetry ⇒ $x_r(t)$ → even $x_i(t)$ → odd
- conjugate asymmetry ⇒ $x_r(t)$ → odd $x_i(t)$ → even

$e^{j\omega t^2}$ → even by no CS/CSS

Periodicity: whose value repeats after period T

- Smallest T such that $x(t+T) = x(t)$ is known as period $\int_a^{a+nT} x(t)dt = \int_b^{b+nt} x(t)\,dt$
- for discrete time signal, period (N) must be an integer.
- All finite length signals are aperiodic
- $x_1(t) \to T_1 \quad x_2(t) \to T_2$

  $\Rightarrow x_1(t) + x_2(t) = z(t)$

  $z(t+T) = z(t) = x_1(t) + x_2(t)$

  $x_1(t+T) + x_2(t+T)$

  $T = nT_1, \; T = mT_2 \quad \frac{T_1}{T_2} = \text{rational} \to \text{periodic}$
- $x_1[n] \to N_1 \quad x_2[n] \to N_2$

$\Rightarrow x_1[n] + x_2[n] \rightarrow N \quad N = LCM(N_1, N_2)$

- scaling / shifting changes period but periodicity remains

Energy of signal:

- $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$ $\longrightarrow$ conservation of energy not valid
- pythagoren relation is valid though
- Signals with finite energy are called Energy signals
- $\int_{-\infty}^{\infty} |x(t) + y(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt + \int_{-\infty}^{\infty} |y(t)|^2 dt + \int_{-\infty}^{\infty} x^*(t) y(t) dt + \int_{-\infty}^{\infty} x(t) y^*(t) dt$

Orthogonal $\Rightarrow$ this sum is 0

- Also $E_{xy} \neq E_x E_y$
- Energy of periodic signals $= \infty$

Power of a Signal:

- $P_x = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$

- Average energy of a signal
- If $E_x < \infty \Rightarrow P_x = 0$ and $P_x > 0 \Rightarrow E_x = \infty$
- signals with finite power ($P_x < \infty$) are called power signals.

Negative frequency: direction of rotation of phasor

- +ve → anticlockwise
- -ve → clockwise

frequency: rate of change of phase

- +ve → increasing phase
- -ve → decreasing phase

- Time shifting doesn't affect Energy / Power
- Time scaling $x(at)$, energy becomes $E_x/|a|$ and power is same

System: A box that processes an input (single) and produces an output (single)

- Integrator: $y(t) = f(x(t)) = \int_{-\infty}^{t} x(\theta)\,d\theta$
- differentiator: $y(t) = f(x(t)) = d/dt\,(x(t))$
- time shifter: $y(t) = x(t-T)$
- amplifier: $y(t) = Ax(t)$

- Causal: Output at time t only depends on the input at time t or earlier
- Non causal: output at time t depends on future as well (can be implemented with delay)
- Memory less: Output at time t depends only on the input at time t
- System with memory: Output at time t depends on the past (or future) inputs
- Non-causal $\Rightarrow$ System with memory
- Integrator, differentiator $\rightarrow$ System with memory

- Invertible system: given an output, the corresponding input can be found
- No memory $\Rightarrow$ causal
- derivative, integrator $\rightarrow$ causal

Stable systems:

- BIBO - bounded input bounded output
- for $|x(t)| < \infty$, a stable system gives an output such that $|f\{x(t)\}| < \infty$
- An invertable system isn't necessarily stable and vice versa ex: $y(t) = (t^2+1)x(t)$; $y(t) = x^2(t)$

Time invariance:

- The system's description (input-output mapping) doesn't change with time
- for time invariant $f\{\}$, if $f\{x(t)\} = y(t)$, then $f\{x(t+T)\} = y(t+T)$

eg: $\int_{-\infty}^{t} x(\tau)\, d\tau$

Time varying systems: systems have different input output mappings at different times

- $f\{\}$, $f\{t, x(t)\} = y(t)$ and $f\{t+T, x(t+T)\} \neq f\{t, x(t)\}$

  ex: $e^{-t}x(t)$

LTI systems:

$$\int \delta(t-\theta)\, x(t)\, dt = x(\theta)$$

$$\Rightarrow x(t) = \int_{-\infty}^{\infty} \delta(\tau - t)\, x(\tau)\, d\tau$$

infinite sum of scaled dirac deltas

$$f\{\delta(t-\tau)\} = h_\tau(t)$$

$$f\{x(\tau)\,\delta(t-\tau)\} = x(\tau)\, h_\tau(t)$$

$$f\left\{\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau\right\} = f\{x(t)\} = \int_{-\infty}^{\infty} x(\tau)\, h_\tau(t)\, d\tau$$

Fourier Analysis :-

Eigenfunctions of LTI systems:

$$h(t) * e^{j\omega t} = \int h(\tau) e^{j\omega(t-\tau)} d\tau = e^{j\omega t} \lambda$$

$$\lambda = \int h(\tau) e^{-j\omega\tau} d\tau$$

Complex exponentials are orthogonal and bounded.

Convolution → scaled impulses

If inputs are eigenfunctions:

$$x[n] = \sum c_m b_m[n]$$

convolution: $x[n] = \sum_{m=-\infty}^{\infty} x[m]\, \delta[n-m]$

what if $b_m[n] = e^{j 2\pi \omega_m n}$

$A\theta = \lambda\theta \quad v_1, v_2, \ldots v_n$

$$x = \sum_{m=1}^{n} c_m \theta_m$$

$$\Rightarrow Ax = \sum \lambda_m c_m \theta_m$$

$x(t) = \sum c_m b_m(t)$ [decomposition]

$z(t) = x(t) - \sum c_m b_m(t)$ we wanna minimize $z(t)$

$z(t) \rightarrow$ error kinda

$$E_z = \int |z(t)|^2 dt = \int \left(x(t) - \sum_{m=1}^{M} c_m b_m(t)\right)^2 dt$$

we wanna minimize E_z

$$\frac{dE_z}{dc_i} = 0$$ (it can be complex too)

$$\Rightarrow \int \left(x(t) - \sum_{m=1}^{M} c_m b_m(t)\right)\left(b_i^*(t)\right) = 0$$

$$\Rightarrow \int x(t) b_i^*(t) - c_i b_i(t) b_i^*(t) - \underbrace{\sum_{m \neq i} c_i b_i(t) b_i^*(t)}_{0 \rightarrow \text{orthogonal basis}} = 0$$

$$c_m = \frac{\langle x(t), b_m(t) \rangle}{\langle b_m(t), b_m(t) \rangle} \Rightarrow E_{err} = \underbrace{E_x - \sum |c_m|^2 E_{basis}}_{\geq 0}$$

BESSEL

basis: $b_m(t) = e^{-j\omega_m t}$

FOURIER

complex functions are eigenfunctions of LTI systems $Re^{j\omega t}$. Instead of taking $x(t)$ as scaled impulses, we can treat them as sum of scaled exponentials

$$x[n] = \sum c_m b_m[n] \qquad b_m[n] = e^{j2\pi\omega_m n}$$

$$\Rightarrow y[n] = \sum_m c_m \lambda_m b_m[n]$$

$A\theta = \lambda\theta \Rightarrow$ All eigenvectors θ form the basis of the vector space we're working with

inner product: $u^T\theta$

$$\lambda_m = \sum_{u=-\infty}^{\infty} e^{-j\omega_m 2\pi u} n[u]$$

$z(t) = x(t) - \sum c_m b_m(t)$

drive to 0

take $c_i = p_i + iq_i$, find p_i, q_i by $dE/dp_i = 0$, $dE/dq_i = 0$

$$c_m = \frac{\langle x(t)\, b_m(t)\rangle}{\langle b_m(t)\cdot b_m(t)\rangle}$$

$$E = \int |z(t)|^2 dt$$

$$\frac{dE}{dc_i} = 0 \Rightarrow c_i = \frac{\int x(t) b_i^*(t)\, dt}{\int b_i(t)\, b_i^*(t)\, dt} \longrightarrow \text{optimal } c_i\text{'s}$$

To make sure energy is finite: (basis are orthogonal)

focus on $t \in [-T/2, T/2]$

$b(t)$ should be periodic with one of its periods T, $x(t)$ should also be periodic with period T.

we can decompose any periadic signal using its sinusoid harmonics.

Restrictions on basis functions:

$$\int_{-T/2}^{T/2} b_i(t)\, b_j^*(t) = 0 \quad \text{for } i \neq j \quad \text{(orthogonality)}$$

Sinusoids we can have are:

$e^{j 2\pi/T\, kt}$ → depends on period of $x(t)$

$$x(t) = \sum_{m=-\infty}^{\infty} c_m e^{j 2\pi m t/T}$$

$$\Rightarrow c_m = \frac{\int_{-T/2}^{T/2} x(t)\, e^{-j 2\pi m t/T}\, dt}{\int_{-T/2}^{T/2} e^{j 2\pi m t/T} \cdot e^{-j 2\pi m t/T}\, dt} = \frac{1}{T} \int_{-T/2}^{T/2} x(t)\, e^{-j\frac{2\pi}{T} m t}\, dt$$

When does fourier representation exist?

$$c_m = \frac{\langle x(t)\, b_m^*(t) \rangle}{\langle b_m(t)\, b_m^*(t) \rangle}$$ to be finite, $\int_{-T_0}^{T_0} x(t)\,dt < \infty$

absolutely integrable

$x(t)$ should have finite number of discontinuities → converges to midpoint by MVT

for finite discontinuities E tends to 0 when integrated

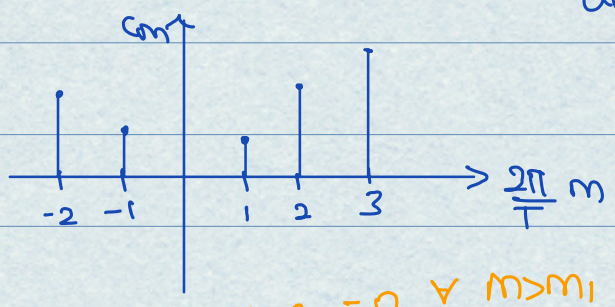
$x(t)$ should have finite number of maximas and minimas.

$E_{x(t) \text{ in one time period}} = T \sum |c_m|^2$ parseuals theorm

$\Rightarrow P = \sum |c_m|^2$

We can represent $\sum_{m=-\infty}^{\infty} c_m e^{j\frac{2\pi m t}{T}}$ in the frequency domain

($\frac{2\pi m}{T}$ → corresponding frequency)



(spectrum)

m_1 s.t $c_m = 0$ $\forall$ $m > m_1$

Bandwidth = $m_1 - m_2$ → m_2 s.t $c_m = 0$ $\forall$ $m < m_2$

3Db bandwidth: range within where frequency falls by $1/2$.

Focussing on a periodic signal within $(-T/2, T/2)$

$$c_m = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-j\omega_m t}\, dt$$

$\omega_m \to m\omega_0$

$dt \to 2\pi f_0 = 2\pi/T_0$

Because $x(t)$ is periodic, all points from $-T/2$ to $T/2$ (uncountable ∞) can be represented using a countably ∞ set.

Real signals: m graph is symmetrical in a way

$$x(t) = \cos(2\pi t) + \sin(10\pi t)$$

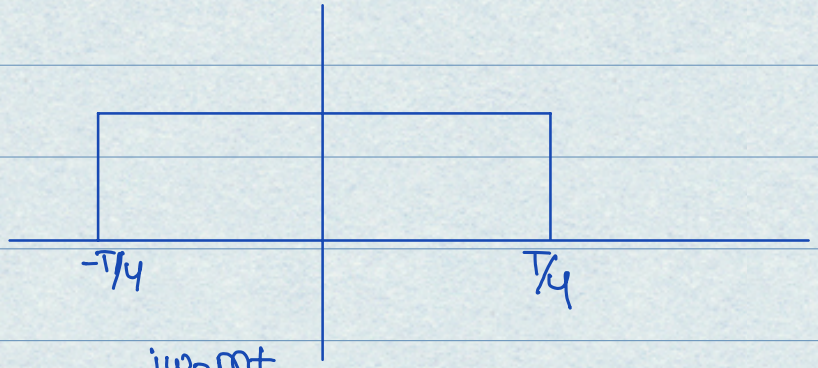
$$\frac{e^{j(2\pi t)} + e^{-j(2\pi t)}}{2} + \frac{e^{j(10\pi t)} - e^{-j(10\pi t)}}{2}$$

period $= 1 \Rightarrow \omega_0 = 2\pi$

$$c_m = \begin{cases} 1/2 & |m| = 1 \\ 1/2j & m = 5 \\ -1/2j & m = -5 \\ 0 & \text{else} \end{cases}$$

$\longrightarrow$ Bandwidth $= 10\pi$ rad/sec

$$x(t) = \sum \text{rect}\left(\frac{2t}{T} - nt\right)$$

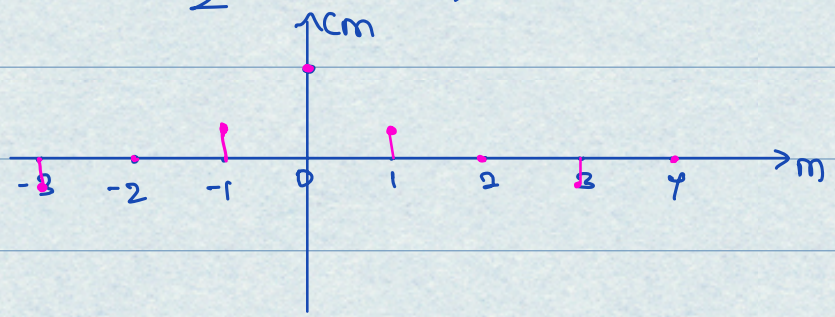


$x(t) = \sum c_m e^{j\omega_0 m t}$ $\quad \omega_0 = 2\pi/T$

$$c_m = \frac{1}{T}\int_{-T/4}^{T/4} e^{-j\omega_0 m t}\,dt = \frac{1}{2}\,\frac{\sin(\omega_0 m T/4)}{\omega_0 m T/4}$$

$$\text{sinc}\,x = \frac{\sin \pi x}{\pi x} = \frac{1}{2}\,\text{sinc}\left(\frac{\omega_0 m T}{4\pi}\right)$$ ∞ bandwidth

$$= \frac{1}{2}\,\text{sinc}(m/2)$$



time domain $x(t)$	spectral domain c_m
even	even
odd	odd
Real	C.S
Imaginary	C.A

In the spectrum when $m=0$

$$C_0 = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\,dt$$ DC component of signal

even signals: $x(t) = C_0 + \sum_{m=1}^{\infty} 2C_m \cos(\omega_0 m t)$

odd signals: $x(t) = C_0 + \sum_{m=1}^{\infty} 2j\, C_m \sin(\omega_0 m t)$

real, odd signals: $C_m = -C_{-m}$

$$C_m^* = C_m$$

real, even signals: $C_m = C_{-m}$

$$C_m^* = C_m$$

$x(t) = \Sigma C_m b_m(t) \Rightarrow x^*(t) = \Sigma d_m b_m(t)$

$\Rightarrow d_m = C_{-m}^*$ no bandwidth change

$\Rightarrow x(-t) = \Sigma e_m b_m(t)$

$\Rightarrow e_m = C_{-m}$ no bandwidth change

$\Rightarrow x(t-\varsigma) = \Sigma f_m b_m(t)$

$f_m = C_m e^{-j\omega_0 m(\varsigma)}$

frequency dependent phase shift

more f_1 more phase shift and same for less

$\Rightarrow x(at) = \sum g_m b_m(t)$

(period $= T/a$)

$\Rightarrow g_m = c_m/a$ → we can do this when a is an integer

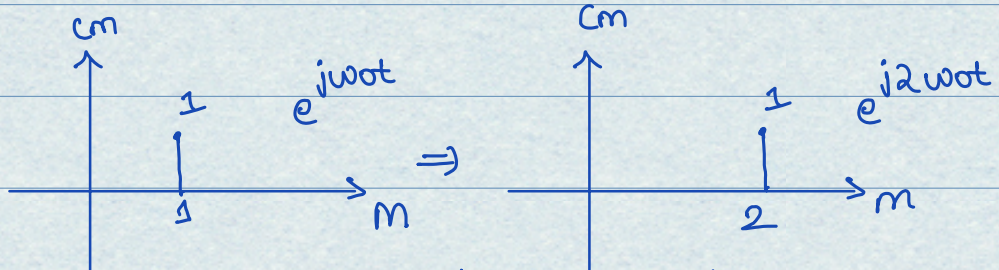
ω_0 changes $\omega_0' = \omega_0/a$

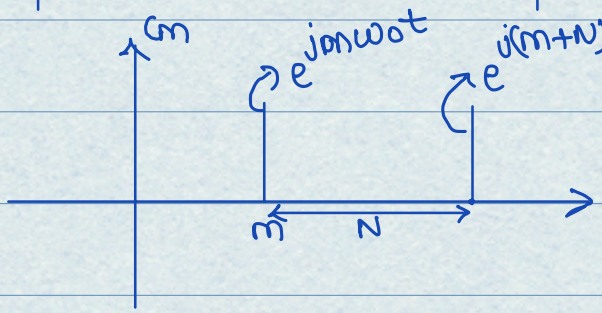
$\Rightarrow$ Spectrum remains the same

$b_m = c^*[m]$

$$\Rightarrow c^*[m] = \frac{1}{T}\int_{-T/2}^{T/2} x^*(t)\, e^{j\omega_0 m t}\, dt$$

$$= \frac{1}{T}\int_{-T/2}^{T/2} x^*(-\theta)\, e^{-j\omega_0 m\theta}\, d\theta$$





$$c_m = \frac{1}{T}\int x(t)\, e^{-j\omega_m t}\, dt$$

$$\Rightarrow c_{m-N} = \frac{1}{T}\int x(t)\, e^{-j\omega(m-N)t}\, dt$$

$$= 1/T \int x(t)\, e^{-j\omega_m t}\cdot e^{+j\omega_0 N t}\, dt$$

$c[m-N] \longrightarrow = x(t) \cdot e^{jw_0 Nt}$

$c[am]$:

$a \rightarrow$ integer $\quad c_m = \frac{1}{T}\int x(t) e^{-jwmt}$

$c_{am} = \frac{1}{T}\int x(t) e^{-jw_0(am)t}$

information is lost, we don't see

$\Rightarrow$ signal itself changes, we cant find relation.

For a signal with periodic impulse response $h(t)$ and periodic input $z(t)$:

$$z(t) = \sum_{k=-\infty}^{\infty} a_k e^{jkwt} \qquad w = \frac{2\pi}{T}$$

$$h(t) = \sum_{m=-\infty}^{\infty} b_m e^{jmw}$$

RC circuit

input $x(t) = \sum c_m e^{jw_0 mt}$ → d_m

$\Rightarrow y(t) = \sum \left(c_m \frac{1}{1 + RCjmw_0} \right) e^{jw_0 mt}$



we can feed in a square wave and get a sinusoid depending on RC values.

$x(t) \longrightarrow c_m$

$y(t) = \frac{d}{dt} x(t) \longrightarrow \sum c_m . jw_0 m \, e^{jw_0 mt}$

$c_m jw_0 m$

(High pass filter) → small w_0 ⇒ higher m values are only see

$y(t) = \int_{-T/2}^{t} x(t)dt \longrightarrow y(m) = \sum c_m \frac{e^{jw_0 mt} - e^{-jw_0 mT/2}}{jw_0 m}$

(low pass filter) $\Rightarrow \frac{c_m}{jw_0 m} - \frac{e^{-jw_0 mT/2}}{jw_0 m}$

fourier series coefficients

additive constant for periodic signal

Find the fourier series of the impulse train:

$$\sum_{n=-\infty}^{\infty} \delta(t-nT)$$

$$C_m = \frac{1}{T}\int_{-T/2}^{T/2} \sum_{n=-\infty}^{\infty} \delta(t-nT)\, e^{-jw_0mt}\, dt$$

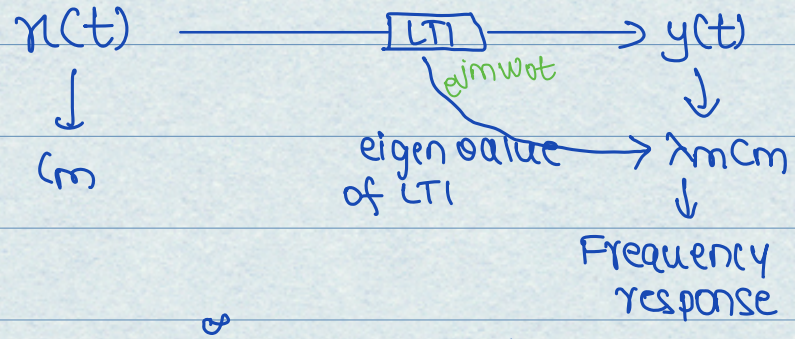
$$C_m = \frac{1}{T}$$

$$x(t) = \sum_{m=-\infty}^{\infty} \frac{1}{T} e^{jw_0mt} \rightarrow \text{used inside analog to digital converter}$$

$$x(t)\left[\sum_{n=-\infty}^{\infty} \delta(t-nTs)\right] \rightarrow \text{sampling signals every } T \text{ seconds}$$

$x(t)$ → T → a_m

$\delta(t-nTs)$ → c_m

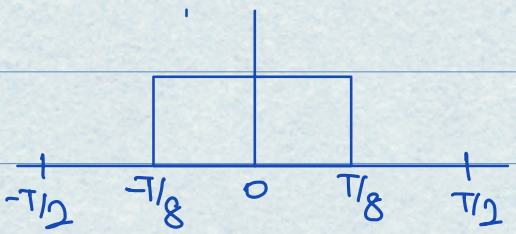


$$\lambda_m = \int_{-\infty}^{\infty} h(t)\, e^{-jmw_0t}\, dt$$

frequency response ↓ ; $h(t)$ → impulse response

Fourier Transform.

$$x(t) = \sum c_m e^{jm\omega_0 t}$$



$$\frac{1}{D}\,\text{Sinc}(K/4)$$

As T increases (D decreases), ω decreases

$\Rightarrow$ adds more frequencies (harmonics)

$$x(t) = \sum \left[1/T \int_{-T/2}^{T/2} x(t)\, e^{-jm\omega_0 t}\, dt \right] e^{jm\omega_0 t}$$

$$\int_{-T/2}^{T/2} x(t)\, e^{-jm\omega_0 t}\, dt = X(m\omega_0)$$

$$x(t) = \sum \frac{1}{T_0} X(m\omega_0)\, e^{jm\omega_0 t}$$

$T_0 \to \infty$
$\Rightarrow \omega_0 \to 0$

$$= \sum \frac{\omega_0}{2\pi} X(m\omega_0)\, e^{jm\omega_0 t}$$

$$= 1/2\pi \lim_{\omega_0 \to 0} \sum_{m=-\infty}^{\infty} \omega_0 X(m\omega_0)\, e^{jm\omega_0 t}$$

$$= \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

$$X(\omega) = \lim_{\substack{\omega_0 \to 0 \\ \text{or} \\ T_0 \to 0}} \int_{-T_0/2}^{T_0/2} x(t)\, e^{-jm\omega_0 t}\, dt = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$\int_{-\infty}^{\infty} e^{j\omega_0 t} e^{-j\omega_1 t}\, dt = \begin{cases} 0 & \omega_0 \neq \omega_1 \\ \infty & \omega_0 = \omega_1 \end{cases}$$

$$= c\,\delta(\omega_0 - \omega_1)$$

$$\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} e^{j\omega t}\, dt \right) d\omega \qquad \text{(for scaling factor)}$$

$$\downarrow$$

$$\lim_{T \to \infty} \int_{-T/2}^{T/2} e^{j\omega t}\, dt = \frac{2}{\omega} \sin \omega T/2 = T \operatorname{sinc}(\omega T/2\pi)$$

$$\downarrow$$

As $T \to \infty \Rightarrow$ looks like dirac

$$\text{area: } \int_{-\infty}^{\infty} \frac{\sin(\omega T/2)}{\omega/2}\, d\omega = 4 \int_{0}^{\infty} \frac{\sin(\omega_0 T)}{\omega_0}\, d\omega$$

* we know that area of normalized sinc(x) is 1

$$\frac{1}{\omega_0} = \int_0^{\infty} e^{-\omega_0 t}\, dt$$

$$\Rightarrow 4 \int_0^{\infty} \int_0^{\infty} \sin(\omega_0 T)\, e^{-\omega_0 t}\, d\omega_0\, dt$$

$$= 4 \int_0^{\infty} \frac{T}{T^2 + t^2}\, dt = 2\pi \rightarrow \text{irrespective of } \omega_0, T$$

$$\Rightarrow \boxed{c = 2\pi}$$

$$x(t) = \int X(\omega)\, e^{j\omega t}\, d\omega$$

inner product

$$\int x(t)\, e^{-j\omega_0 t}\, dt = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t}\, d\omega \right) e^{-j\omega_0 t}\, dt$$

$$= \int_{-\infty}^{\infty} X(\omega) \left(\int_{-\infty}^{\infty} e^{j(\omega-\omega_0)t}\, dt \right) d\omega$$

$$= 2\pi X(\omega_0) \qquad 2\pi\,\delta(\omega-\omega_0)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \longrightarrow \text{fourier transform}$$

↳ spectral density → for continuous functions

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega \longrightarrow \text{inverse fourier transform}$$

parseval's theorm: $E_x = \int |x(t)|^2 dt$

$$= \frac{1}{2\pi} \int |X(j\omega)|^2 d\omega = \frac{E_x}{2\pi}$$

$$\int |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t)\, x^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega \right) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} x^*(j\omega)\, e^{-j\omega t}\, d\omega \right) dt$$

$$= \int \frac{1}{4\pi^2} \iint X(j\omega)\, X^*(j\theta)\, e^{j(\omega-\theta)t}\, d\omega\, d\theta\, dt$$

$$= \frac{1}{4\pi^2} \iint X(j\omega)\, X^*(j\theta) \underbrace{\int e^{j(\omega-\theta)t}\, dt}_{2\pi\delta(\omega-\theta)}\, d\omega\, d\theta$$

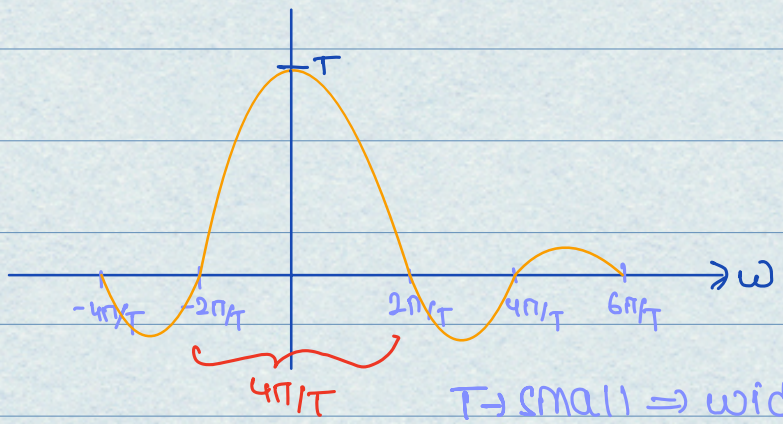
$$= \frac{1}{2\pi} \int X(j\omega)\, X^*(j\omega)\, \delta(\omega-\theta)\, d\omega\, d\theta$$

$$= \frac{1}{2\pi} \int |X(j\omega)|^2 d\omega = \frac{E_x}{2\pi}$$

$$x(t) = \text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1 & t \in [-T/2, T/2] \\ 0 & \text{else} \end{cases}$$

$$F.T = X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{j\omega t}\, dt = \int_{-T/2}^{T/2} e^{-j\omega t} = \frac{2\sin(\omega T/2)}{\omega}$$

$$= T\,\text{sinc}\left(\frac{\omega T}{2\pi}\right)$$



$T \to \text{small} \Rightarrow \text{width} \to \infty$

$T \to \text{large} \Rightarrow \text{width} \to 0$

$$\text{signal duration} \propto \frac{1}{\text{Bandwidth}}$$

uncertainity principle

$x(t)$ → time domain	$X(j\omega)$ → frequency domain	Bandwidth → h
$x^*(t)$	$X^*(-j\omega)$	No change
$x(-t)$	$X(-j\omega)$	No change
$x(t-s)$	$X(j\omega)e^{-j\omega s}$	No change
$x(t)e^{j\theta t}$	$X(j(\omega-\theta))$	No change
$x(ct)$	$X(j\omega/c)/\lvert c\rvert$	$c>1$ → expand $c<1$ → shrink

Time	Freq
$x(t)\cdot y(t)$	$X(j\omega)^* Y(j\omega)/2\pi$
$x(t)^* y(t)$	$X(j\omega)Y(j\omega)$

Fourier transform of $u(t)$

$$X(\omega)=\int_{-\infty}^{\infty} u(t)\,e^{-j\omega t}=\int_{0}^{\infty} e^{-j\omega t}=\frac{e^{-j\omega t}}{j\omega}\Big|_0^{\infty}$$

we dont know where its converging

instead, $u(t) = \lim_{a \to 0} e^{-at} u(t)$ → decays as $t \uparrow$

$$X(\omega) = \frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \Big|_0^\infty = \frac{1}{a+j\omega}$$

$$\lim_{a \to 0} X(\omega) = \lim_{a \to 0} \frac{1}{a+j\omega} = \frac{1}{j\omega} \quad (\text{as long as } \omega \neq 0)$$

$$X(\omega) = \frac{1}{j\omega} + C\delta(\omega)$$

$$\mathcal{F}\{1\} = 2\pi\delta(\omega) \Rightarrow \mathcal{F}\{u(t) + u(-t)\} = 2\pi\delta(\omega)$$

$$\Rightarrow \frac{X(\omega) + X(-\omega)}{2} = \pi\delta(\omega)$$

$$\Rightarrow \boxed{C = \pi}$$

Fourier Transform:

$$z(t) = \sum \frac{1}{T} \left(\int_T z(t) e^{-jk\omega t} dt \right) e^{jk\omega t}$$

$$= \sum \frac{\omega}{2\pi} Z(jk\omega) e^{jk\omega t}$$

$$\lim_{T \to \infty} z(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Z(j\nu) e^{j\nu t} dt$$

spectrum becomes dense

Energy spectral density: $|z(j\omega)|^2$

$$z(jk\omega) = \int_T z(t) e^{-jk\omega t}$$